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Bachelor's Thesis

Measuring Drift in Therapeutic AI: A Stability-Based Evaluation of Sovereign LLMs in Nightmare Therapy

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Declaration of Authorship

The content of this thesis represents my own knowledge, my own understanding and my own perspective on the topic. In case artificial intelligence tools were used, their way and purpose of usage has been made transparent. Moreover, I have cited all my sources in accordance with academic standards. I am ready and able to explain and defend the positions developed in this thesis. This thesis has not been submitted, either in part or whole, at this or any other university.

Osnabrück, 31.03.2026

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Abstract

Large language models are increasingly used as therapeutic agents in mental health applications, but it is unclear how consistently they respond when given the same patient input multiple times. This thesis measures therapeutic stability in the rescripting phase of Imagery Rehearsal Therapy, where the model must choose a therapeutic strategy and generate a response that carries it out.

Ten models, including the EU-sovereign Mistral Large 3, two proprietary models (GPT-5.4, Claude Sonnet 4.6), and seven open-weight comparators, are evaluated across 6,000 trials at five temperature settings. A three-level evaluation framework compares strategy selections (Jaccard similarity), response texts (BERTScore), and whether the model does what it declared (LLM judge coherence).

The main findings are: (1) Mistral Large 3 matches the proprietary models on plan consistency at its recommended temperature range. (2) Temperature affects which strategies the model picks but not how well it carries them out. The instability sits in strategy selection, not in response quality. (3) Greedy decoding ($T = 0.0$) is not optimal for all models; three achieve higher consistency at $T = 0.075$. (4) BERTScore captures whether the therapeutic tone stays similar but does not detect changes in the underlying strategy. The three levels together reveal patterns that no single metric shows on its own.

At low temperatures, the EU-sovereign Mistral Large 3 reaches the same plan consistency as the proprietary models. At higher temperatures, the proprietary models remain stable while Mistral degrades. The framework itself is not specific to IRT and could be applied to other structured therapy protocols.

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1. Introduction

1.1. Nightmare Disorder and Imagery Rehearsal Therapy

Recurring nightmares affect an estimated 2–5% of adults (Li et al., 2010; Sandman et al., 2013; Schredl, 2010) and frequently co-occur with post-traumatic stress disorder (PTSD) and depression (Morgenthaler et al., 2018). The recommended treatment, Imagery Rehearsal Therapy (IRT), follows a three-stage protocol: the patient records a nightmare, rescripts it with the therapist, and rehearses the new version (Krakow & Zadra, 2006). A persistent shortage of trained therapists limits its availability (Wittchen et al., 2011), making scalable, AI-assisted delivery a practical goal.

However, the rescripting step requires active strategy decisions from the therapist. An AI agent that selects different strategies on repeated runs of the same case is clinically unreliable, regardless of how capable any single response may be. Patients also differ in how they present: an anxious patient and a resistant patient pose different challenges, and a model that is stable for one profile may not be stable for another.

1.2. AI-Assisted Psychotherapy

Large language models are increasingly deployed in mental health, from empathy-based conversational tools to protocol-driven agents that deliver structured interventions such as cognitive behavioural therapy (Stade et al., 2024; Q. Zhang et al., 2025). reDreamAI is the deployment context for this study: a consumer-facing IRT chatbot that guides users through all stages of the protocol. It is not a research prototype but a real deployment target where reliability is non-negotiable.

This deployment context creates a specific problem. A general-purpose assistant has no therapeutic obligation. Response variability is acceptable and even desirable. A protocol-driven agent, by contrast, must make consistent strategic decisions. If the model selects different rescripting strategies for the same patient on different runs, the patient receives inconsistent care. Unlike rule-based chatbots whose responses can be reviewed once before deployment (Fitzpatrick et al., 2017), an LLM agent generates a new response on every run. Each run is a new draw from the output distribution, so stability must be evaluated repeatedly rather than verified once.

IRT targets nightmares rather than psychiatric diagnoses, which lowers the clinical risk relative to full psychotherapy. Ethical considerations are discussed in section 6.5 and section 7.2. The EU AI Act classifies autonomous therapeutic AI as high-risk and requires auditability (Aboy et al., 2024; van Kolfschooten & van Oirschot, 2024). Data sovereignty requirements push toward EU-resident models (Dale, 2025), introducing a constraint on model selection discussed in section 3.8.

1.3. The Evaluation Gap

Standard NLP benchmarks measure what a model *can* do, not whether it does the same thing reliably. A model can score well on a single evaluation while producing different therapeutic strategies on repeated runs of the same input. Capability and stability are distinct properties, but current benchmarks conflate them.

The closest existing work is BOLT (Chiu et al., 2024), which evaluates LLM therapist behaviour using 13 behavioural codes from motivational interviewing. BOLT measures whether a model applies appropriate therapeutic techniques (quality). This study measures a different property: whether a model applies the *same* techniques across repeated runs (stability). BOLT evaluates single responses; this study evaluates the distribution of responses across twenty independent trials per condition.

Other therapeutic AI evaluations are qualitative, post-hoc, or single-session (Talat et al., 2024). They cannot detect cross-run inconsistency because they do not present the same input multiple times. Systematic, reproducible in silico trials offer a way to measure stability at scale without requiring human participants.

1.4. Research Objectives

This thesis addresses the following research question:

> How consistent are LLM therapeutic responses across independent runs when given the same patient context within a structured IRT protocol?

Two contributions follow from this question.

Contribution 1. A three-level evaluation framework that measures stability at each layer: *plan consistency* (do strategy selections repeat?), *response similarity* (do the generated texts convey the same meaning?), and *plan-response coherence* (does the model do what it declared?). Together, these layers reveal divergence patterns that no single metric can detect.

Contribution 2. A cross-model comparison that applies this framework to sovereign, proprietary, and open-weight models under controlled temperature conditions.

The study isolates the IRT rescripting phase because it is the stage that requires active strategy decisions. Isolating a single stage eliminates stage-routing confounds and allows any measured instability to be attributed to the therapist model rather than to upstream variation in the conversation. chapter 3 describes the experimental design in detail.

2. Background

2.1. Imagery Rehearsal Therapy

Nightmare disorder affects an estimated 2–5% of the adult population (Li et al., 2010; Sandman et al., 2013; Schredl, 2010) and shows strong comorbidity with Post-Traumatic Stress Disorder (PTSD) and depression (Morgenthaler et al., 2018). For patients with recurrent nightmares, the distress extends beyond the dream itself: sleep avoidance, daytime fatigue, and heightened anxiety create a cycle that standard sleep hygiene cannot break. IRT is the recommended treatment for this condition (Krakow & Zadra, 2006; Morgenthaler et al., 2018).

IRT follows a structured three-stage protocol. In the *recording* stage, the patient describes the nightmare in detail. In the *rescripting* stage, the patient and therapist collaboratively transform the nightmare narrative to reduce distress and increase mastery. In the *rehearsal* stage, the new narrative is consolidated and the patient prepares to rehearse it before sleep. Krakow and Zadra (2006) describe the rescripting instruction as deliberately open-ended: patients are told to "change the nightmare any way you wish."

The rescripting stage is the cognitive core of the protocol. Unlike recording (which requires listening) or rehearsal (which requires repetition), rescripting demands that the therapist choose a specific rescripting strategy and guide the patient through it. These choices determine the character of the intervention. A therapist who helps the patient confront a nightmare threat delivers a different treatment from one who introduces a protective figure or reframes the threat as harmless.

Clinical research has identified distinct categories of rescripting strategies. Harb et al. (2012) coded five types from combat veteran rescripts: alternative endings (58%), positive image insertion (23%), threat transformation (13%), reality markers (10%), and distancing (8%). Germain et al. (2004) proposed a complementary classification through the Multidimensional Mastery Scale, which groups rescripting outcomes along physical, social, environmental, emotional, mystical, and avoidance dimensions. Both frameworks show that rescripting is not a single technique but a family of distinct interventions. Which strategy the therapist selects is therefore a meaningful clinical decision, and the consistency of that decision across sessions is what this thesis measures.

Treatment fidelity instruments provide the clinical standard for measuring whether a therapist delivers an intervention correctly. The IRT literature draws on two established frameworks: the ENACT rating scale (Kohrt et al., 2015) and the NIH BCC recommendations (Bellg et al., 2004). Both grade therapist adherence on a ternary scale: *absent*, *partial*, or *implemented*. This gradation captures clinically meaningful differences. A therapist who briefly mentions a rescripting strategy without developing it (partial) delivers a different intervention from one who fully guides the patient through it (implemented).

This ternary framework provides a scoring foundation that transfers directly to computational evaluation (see subsection 3.7.3). However, existing fidelity instruments assume a human clinician. They have not been adapted for Large Language Model (LLM)-based agents, where the therapist is a probabilistic text generator rather than a trained professional.

2.2. Treatment Fidelity in LLM-Based Interventions

The deployment of LLMs in mental health has expanded from empathy tools and symptom screeners to protocol-driven therapeutic agents. Stade et al. (2024) survey this trajectory and argue that LLMs could fundamentally reshape behavioural healthcare delivery, from diagnostic support to full psychotherapeutic interventions. Early systems such as WOEBOT (Fitzpatrick et al., 2017) delivered Cognitive Behavioral Therapy (CBT) through rule-based dialogue trees where every response was pre-authored and reviewed before deployment. For these systems, treatment fidelity is a design property: the chatbot cannot deviate from its script.

LLM-based agents operate differently. Each response is a fresh sample from the model's output distribution. A single model can produce high-quality, protocol-adherent responses on one run and strategically inconsistent ones on the next. This distinction is central to the treatment fidelity literature. Mowbray et al. (2003) define fidelity as the extent to which an intervention follows its intended protocol, covering dimensions such as adherence (doing the prescribed things) and quality of delivery (doing them consistently). For rule-based chatbots, adherence is built in by construction. For LLM-based protocol agents, adherence becomes probabilistic and must be measured across runs rather than assumed from a single review.

LLM-based agents also face general failure modes such as hallucination, stage confusion, and persona inconsistency. These are well-documented concerns for clinical AI but fall outside the scope of this study. The present work focuses on a narrower problem: the probabilistic nature of LLM outputs means that identical therapeutic contexts can produce different outputs on each run. Three types of inconsistency can occur, each with different clinical implications.

First, the model may select different rescripting strategies across runs. As established in section 2.1, strategy selection is a meaningful clinical decision. A patient who receives confrontation-based rescripting in one session and cognitive reframing in the next encounters two different interventions. In a protocol where the therapist is expected to follow consistent principles, this is a fidelity problem in the sense of Mowbray et al. (2003): the model fails to adhere to a stable treatment approach.

Second, even when the model selects the same strategy, the therapeutic language may vary between runs. Some variation is expected and acceptable: a competent human therapist naturally paraphrases while maintaining the same strategic intent. But large variation in wording, even under the same strategy, could indicate unstable delivery. Mowbray's fidelity framework motivates this concern: if consistent delivery matters for human therapists, it matters at least as much for LLM agents whose outputs vary with every run. This study does not measure fidelity to a clinical standard directly. Instead, it measures *consistency across runs*: does the model plan the same techniques each time (Method 1), does it word the response similarly (Method 2), and does it carry out what it declared (Method 3).

Third, the model may declare one strategy but carry out another. If a model states that it will use confrontation but actually delivers a calming sensory exercise, there is a mismatch between intent and execution. This internal incoherence is distinct from the first two types: the model is neither selecting a wrong strategy nor varying its language, but failing to do what it said it would do.

Each type of inconsistency requires a different measurement approach. chapter 3 develops one method for each.

Regulatory frameworks reinforce the need for stability measurement. The European Union (EU) AI Act classifies autonomous therapeutic AI as high-risk, imposing requirements for auditability, documentation, and risk management (Aboy et al., 2024; van Kolfschooten & van Oirschot, 2024). The Medical Device Regulation adds further constraints on software used in clinical contexts. For EU-based deployment scenarios, data sovereignty requirements push toward self-hosted, EU-resident models (Dale, 2025), constraining the model selection space (see section 3.8).

Despite these requirements, stability remains largely unmeasured. Prior therapeutic AI evaluation is qualitative, post-hoc, or single-session. Chiu et al. (2024) represent the closest prior work with their BOLT framework, which assesses LLM therapist behaviour across 13 behavioural codes drawn from motivational interviewing. BOLT asks whether the model does the right thing. The present study asks a different question: whether the model does the *same* thing across repeated runs. Existing frameworks, including BOLT, evaluate single responses rather than the distribution of responses across repeated trials.

2.3. Stochastic Behaviour in Language Models

At temperatures above zero, LLM token selection is probabilistic. The softmax distribution over the vocabulary is divided by the temperature parameter before sampling, so higher temperatures flatten the distribution and lower temperatures sharpen it toward the most likely token. At $T = 0$, the model selects the highest-probability token deterministically (greedy decoding). At $T > 0$, tokens are sampled from the resulting distribution, introducing variance into the output.

This per-token variance compounds through autoregressive generation. Each token is conditioned on all preceding tokens, so a single divergent token early in a response can cascade into a different therapeutic strategy. A model that selects "confront" rather than "imagine" at one token position may generate an entirely different rescripting approach in the subsequent response. The clinical impact of stochastic sampling is therefore not proportional to per-token entropy but can amplify through the response.

The clinical consequence is concrete: the same patient presenting the same nightmare on two different days may receive strategically inconsistent care. In a general-purpose assistant, this variability is benign or even desirable. In a structured therapy protocol like IRT, where the therapist is expected to apply consistent techniques across sessions, it maps directly to the three types of inconsistency described in section 2.2: variable strategy selection, variable response wording, and plan-response mismatch.

Prior work on LLM stability addresses different questions than therapeutic consistency. Novikova et al. (2025) survey the consistency landscape, covering whether models give the same answer to the same question across different phrasings and contexts. Xu et al. (2025) measure the opposite problem: LLMs generating repetitive plot structures in creative writing, producing too little variation rather than too much. Blackwell et al. (2024) focus on benchmark reproducibility, showing that evaluation scores fluctuate across repeated runs even with fixed seeds and $T = 0.0$. These lines of research share the concern for cross-run variation but none of them ask whether a model makes the same *clinical decisions* when presented with identical patient contexts.

The mechanical source of instability is well understood. What remains unquantified is its impact on structured therapeutic decision-making. How much does temperature-driven variance cost in clinical consistency? This question motivates the temperature sweep in the experimental design (section 3.9), which tests five points from $T = 0.0$ (deterministic) to $T = 0.6$ (typical vendor default for open-weight models).

2.4. Evaluation Metrics for Text Generation

Measuring consistency between text outputs requires a similarity metric. The choice of metric determines what kind of consistency is captured, and standard options fail in different ways for therapeutic language.

Surface-level metrics such as BLEU and ROUGE measure token overlap between texts. BLEU counts how many short word sequences (n-grams) in one text also appear in the other. ROUGE does the same in the opposite direction, measuring recall rather than precision. Both produce a score between 0 and 1 based on how many tokens match.

In therapeutic contexts, token overlap produces misleading results in both directions. Two responses that use identical words to describe different strategies score high despite being clinically divergent. Conversely, two responses that implement the same strategy through different phrasing score low despite being therapeutically equivalent. Therapeutic paraphrasing is the norm: a competent therapist naturally varies their language while maintaining the same strategic intent. Surface metrics penalise this valid variation.

BERTScore (T. Zhang et al., 2020) addresses this limitation by computing similarity in contextual embedding space rather than token space. Each token in the candidate text is matched to the most similar token in the reference text using cosine similarity over contextualised embeddings, producing precision, recall, and F1 scores. The metric captures semantic similarity independent of surface form: two sentences that mean the same thing but use different words receive high scores.

The choice of embedding model affects BERTScore's behaviour. Among more than 130 available models, those fine-tuned on Natural Language Inference (NLI) tasks achieve the highest correlation with human judgements of semantic similarity. The standard validation method is to compare BERTScore rankings against human similarity ratings from the WMT shared translation tasks, where professional annotators judge whether two texts convey the same meaning. DeBERTa-XLarge-MNLI (He et al., 2021) achieves the highest WMT16 human correlation ($r = 0.778$) among available models. This means it captures most of what human annotators consider semantically similar, though the correlation is not perfect. For therapeutic language specifically, no equivalent human validation exists. The model is selected for Method 2 (subsection 3.7.2) based on its general-domain performance, with the caveat that its sensitivity to clinically meaningful differences has not been independently verified.

BERTScore captures semantic similarity but cannot assess strategic intent. A metric that reports high similarity between two responses does not reveal whether both responses implement the same therapeutic strategy. The LLM-as-judge paradigm addresses this gap. By instructing a separate LLM to evaluate whether a response implements a declared strategy, it becomes possible to assess criteria that no scalar metric can capture: "does this response implement a confrontation strategy?" requires understanding both the strategy definition and the response content.

LLM-as-judge approaches carry known limitations. The judge's scores are not ground truth but a computational estimate. The judge model brings its own biases: self-preference (rating outputs from its own family more favourably), position bias (higher scores for responses presented first), and verbosity bias (rewarding longer responses regardless of quality) (Zheng et al., 2023). These can be mitigated through cross-model judging, structured rubrics, and explicit justification requirements, but they cannot be eliminated entirely. Any study using an LLM judge must treat its scores as approximate and, where possible, validate them against human ratings. The mitigation and validation strategy adopted in this study is detailed in subsection 3.7.3.

No single metric covers what therapeutic stability requires. Embedding similarity captures semantic similarity but misses strategic intent. LLM judges can assess intent but introduce their own reliability concerns. Surface metrics capture neither. Existing therapeutic AI evaluations rely on a single metric type, leaving the gap between semantic equivalence and strategic intent unaddressed. This motivates the three-method design presented in chapter 3.

3. Methods

3.1. System Overview and Design Rationale

The evaluation system separates dialogue creation from stability measurement through two decoupled stacks. The *generation stack* produces synthetic therapy sessions by running a multi-agent dialogue loop. The *evaluation stack* measures the stability of therapeutic responses by presenting frozen conversation histories to the model under test and collecting multiple independent outputs.

This separation is a deliberate design choice. If the evaluation stack generated its own dialogue context, any measured instability could reflect upstream randomness in the patient or routing agents rather than genuine variation in the therapist model. By freezing the conversation history before evaluation begins, the system ensures that every trial receives identical input. Any observed inconsistency is attributable solely to the model under test.

The end-to-end data flow proceeds in three stages: patient vignettes feed the generation stack, which produces complete therapy sessions. These sessions are sliced at rescripting-turn boundaries to create frozen histories. The evaluation stack presents each frozen history to the therapist model across multiple independent trials, collecting both a strategy declaration and a therapeutic response per trial. Finally, an aggregation pipeline aligns all metrics across conditions for comparative analysis.

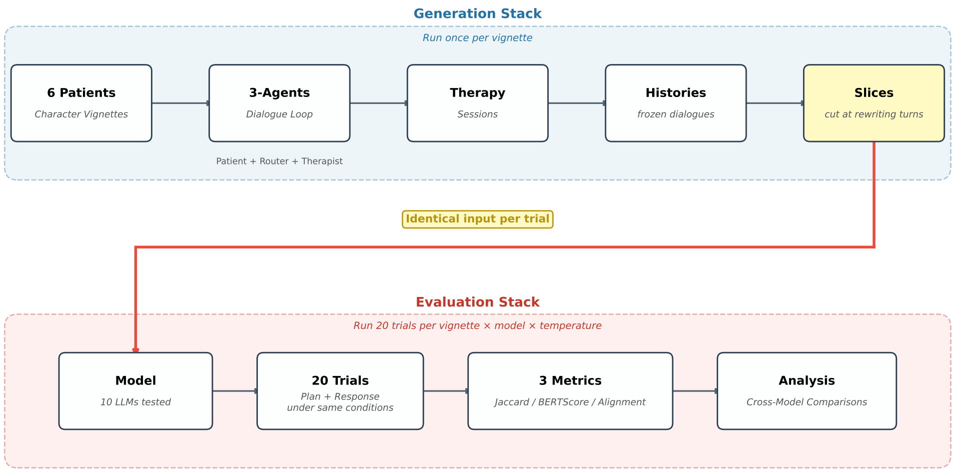

Figure 3.1.: Two-stack pipeline. The generation stack produces frozen conversation histories (run once). The evaluation stack presents each history to the model under test across twenty independent trials per condition. The stacks are decoupled so that evaluation-side variance cannot be contaminated by generation-side randomness.

3.2. Dialogue Generation

The generation stack orchestrates a three-agent loop that produces naturalistic IRT sessions. Each turn proceeds through three agents in sequence:

1. The **patient agent** simulates a clinical presentation based on a Brief Dreamer Interview (BDI)-profiled vignette. Each vignette defines the patient's nightmare content, emotional presentation style, and engagement pattern.

2. The **router agent** classifies the current IRT stage from the conversation history, determining whether the session is in the recording, rescripting, or rehearsal phase.

3. The **therapist agent** generates a stage-appropriate therapeutic response, guided by the router's classification and a stage-specific system prompt.

This three-agent architecture separates the therapist's strategy decisions from stage classification and patient simulation. The router's stage classification controls which system prompt the therapist receives, ensuring that responses are appropriate to the current protocol phase.

The vignette design is informed by anonymised session data from a prior reDreamAI efficacy study. Because data protection requirements prevent using real patient transcripts directly, synthetic patient profiles were created and validated against published clinical research on simulated patients.

Six vignettes cover the range of clinical presentation difficulty expected in IRT deployment: *anxious*, *avoidant*, *cooperative*, *resistant*, *skeptic*, and *trauma*. These profiles range from a cooperative patient who engages readily with rescripting instructions to a trauma patient whose nightmare content triggers intense emotional responses during the session. The design draws on prior synthetic patient work. Z. Wang et al. (2024) demonstrate that expert-defined behavioural principles produce clinically plausible simulated patients. R. Wang et al. (2024) provide cognitive conceptualisation diagrams and conversational styles (direct, resistant, expressive, minimal, deviating, agreeable) that parallel the six profiles used here. Six profiles sample the difficulty range but do not claim exhaustive coverage of patient variability (see section 7.2).

The generation stack traverses the full three-stage IRT protocol, producing complete therapy sessions that are then sliced to create evaluation entry points. Figure 3.2 illustrates this process: the dialogue loops through recording and rescripting until each stage is complete, then proceeds to rehearsal. Frozen history slices are cut from the rescripting stage.

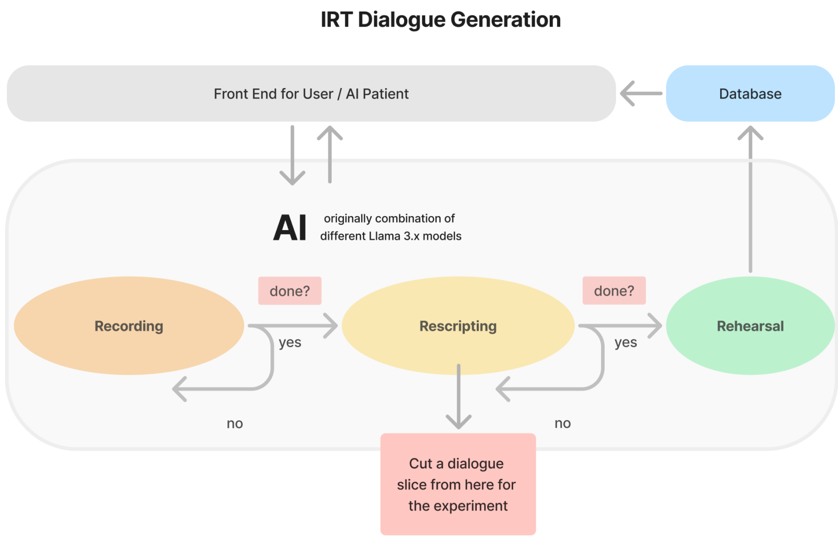

Figure 3.2.: Dialogue generation loop. The three-agent system traverses recording, rescripting, and rehearsal. Frozen history slices for the experiment are cut at rescripting-turn boundaries.

3.3. Frozen History Design

Frozen conversation histories serve as deterministic evaluation entry points. Each vignette's complete therapy session is sliced at rescripting-turn boundaries, where each slice ends after a complete therapist-patient exchange. Every trial, across all models and temperatures, receives the identical frozen history as input. This design eliminates context variation as a confound: if two trials produce different therapeutic responses, the difference is attributable to the model's stochastic sampling, not to differences in the input conversation.

A preliminary analysis tested Mistral Large 3 at five slice depths across all six vignettes at $T = 0.075$ and $T = 0.15$. Jaccard consistency was lowest at the second slice (mean $J = 0.77$ at $T = 0.075$) and rose to near-ceiling values from the third slice onward ($J > 0.93$). The main experiment evaluates at the second rescripting-turn boundary (`slice_2`), making the measured stability a conservative estimate. Slice depth effects are reported in section 5.6.

3.4. Evaluation Stack

The evaluation stack bypasses the router and injects the rescripting prompt directly into the therapist model. Because the IRT stage is known a priori (all frozen histories are sliced within the rescripting phase), routing is unnecessary and would introduce an additional source of variation. Bypassing it isolates the therapist model from stage-classification error.

Each evaluation trial uses *fused generation*: the model produces a `<plan>` block declaring one to two therapeutic strategies, followed by the therapeutic response, in a single call. The plan tokens condition the response tokens in a Chain-of-Thought (CoT)-style generation, so the response is directly conditioned on the declared strategy. This fused approach enables the coherence analysis in Method 3 (subsection 3.7.3): because plan and response are generated in a single pass, any mismatch between declared intent and executed strategy reflects the model's own inconsistency rather than a decoupling artefact. Figure 3.3 illustrates this measurement design.

Twenty independent trials are collected per condition, yielding $\binom{20}{2} = 190$ pairwise comparisons. This trial count balances statistical power with compute cost: 190 pairs provide robust coverage of the pairwise similarity distribution while keeping the total experiment within budget.

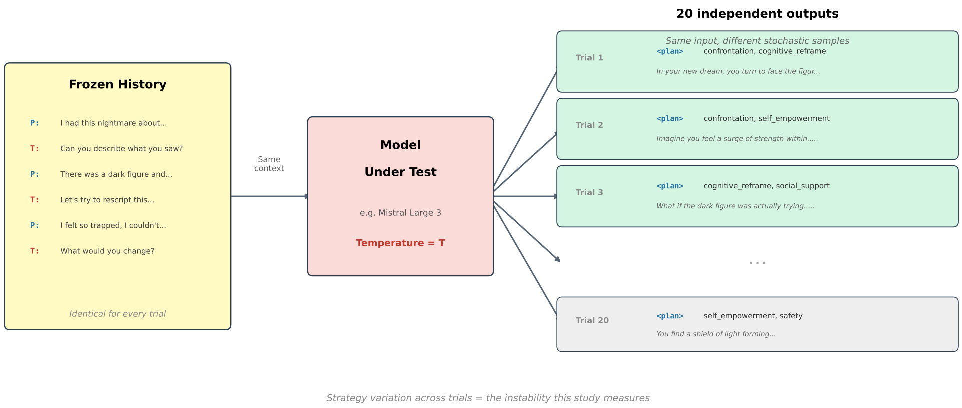

Figure 3.3.: Measurement design. The same frozen history is presented to the model under test twenty times. Each trial independently produces a plan and a therapeutic response. Variation across trials is the instability this study measures.

3.5. The Plan Mechanism

The `<plan>` block raises a methodological question. CoT prompting (Wei et al., 2022) improves output structure by asking models to show their reasoning before answering. However, the faithfulness literature demonstrates that CoT explanations often do not reflect the model's actual computation. Turpin et al. (2023) show that biasing cues in the prompt alter model behaviour without appearing in the CoT trace. Lanham et al. (2023) find that adding mistakes to or paraphrasing the CoT often does not change the final answer, suggesting that models do not always rely on their stated reasoning.

If a model declares its therapeutic strategy before responding, that declaration may be a post-hoc rationalisation rather than a window into its decision-making process. This study resolves the tension by framing the `<plan>` block as *declared intent* rather than a reasoning trace. The plan captures what the model commits to, not what it internally computes. Whether or not the declaration reflects genuine reasoning, a practical consequence holds: if strategy declarations vary across runs, clinical responses will also vary. Variable declared intent predicts variable therapeutic behaviour, which is the clinically relevant property.

This framing sidesteps the faithfulness debate entirely while preserving three concrete uses. First, the declared strategies enable strategy-level consistency scoring (Method 1, subsection 3.7.1). Second, comparison between declared strategies and the executed response enables coherence verification (Method 3, subsection 3.7.3). Third, the plan provides a human-readable summary of the model's intended approach, supporting clinical oversight in deployment.

3.6. Strategy Taxonomy Development

A fixed taxonomy constrains the `<plan>` block to a predefined set of therapeutic strategies. This constraint is necessary for quantitative evaluation: free-text plans cannot be aggregated into set-similarity scores. The taxonomy evolved through four iterations to resolve a severe distribution skew that rendered early versions unusable.

The initial taxonomy (v1) defined eight categories including `empowerment` and `mastery`. Across 1,272 strategy picks, these two categories consumed 87.7% of all selections, while three categories (`social_support`, `sensory_modulation`, and `gradual_exposure`) were never chosen. The root cause was a goal-mechanism conflation: both `empowerment` and `mastery` describe a therapeutic *goal* (the patient gains control) rather than a specific *mechanism* of change. Since every nightmare rescripting intervention serves the goal of increased mastery, the model defaulted to these labels regardless of the actual strategy employed.

Merging the two categories into a single `agency` category (v2) amplified the problem: the merged category achieved 100% dominance. A single super-category that applies to every rescripting scenario provides no discriminative signal.

The resolution in v3 was to split `agency` along the mechanism dimension. `Confrontation` describes external action toward the threat (the dreamer fights, challenges, or resists). `Self_empowerment` describes internal transformation within the dreamer (gaining abilities, growing stronger). These two categories are orthogonal: a dreamer who confronts a monster uses a different mechanism from one who discovers inner strength, even though both gain mastery in the outcome.

A final revision (v4) merged `emotional_regulation` into `sensory_modulation` after smoke testing showed that `emotional_regulation` appeared in only 1 of 120 trials (0.4%). In single-turn rescripting, internal calming manifests as sensory change (the dreamer feels calm, the environment becomes peaceful), making the boundary between the two categories clinically indistinct at this granularity.

The final six-category taxonomy captures concrete therapeutic mechanisms:

The six categories operationalise the clinical rescripting strategies introduced in section 2.1. Avoidance and distancing are excluded because they do not resolve the nightmare threat. **34DFEBDG<empty citation>** review that IRT works

Table 3.1.: Strategy taxonomy (v4). Each category describes a rescripting mechanism rather than a therapeutic goal.

Category	Mechanism
`confrontation`	The dreamer directly faces, fights, or stands up to the threat
`self_empowerment`	The dreamer gains a new ability or inner strength
`safety`	A physical barrier, shelter, or protective figure shields the dreamer
`cognitive_reframe`	Existing dream elements change meaning
`social_support`	A new person, companion, or ally is introduced
`sensory_modulation`	Sensory qualities shift, including internal somatic calming

by increasing mastery over the nightmare, and that avoidance reinforces the fear memory rather than overwriting it. Harb et al. (2012) found that distancing was the least-used strategy (8%) and that violence in rescripted dreams predicted worse outcomes. Reality markers fall outside single-turn rescripting.

Taxonomy design directly shapes measured consistency. A coarse taxonomy inflates agreement (every trial lands in the same category), a fine one deflates it. The sensitivity of scores to category boundaries is revisited in the discussion (section 6.5).

3.7. Evaluation Metrics

Three evaluation methods target distinct layers of the generation process. No single metric captures the full picture of therapeutic stability (section 2.4). Method 1 asks whether the model makes consistent therapeutic decisions. Method 2 asks whether it produces consistent therapeutic responses. Method 3 asks whether it does what it says it will do.

3.7.1. Method 1: Plan Consistency

Method 1 measures whether the model selects the same therapeutic strategies across independent trials. Strategy sets are extracted from the `<plan>` block of each trial output using the fixed taxonomy from Table 3.1. Pairwise Jaccard similarity quantifies agreement between strategy sets:

$$J(A, B) = \frac{|A \cap B|}{|A \cup B|} \tag{3.1}$$

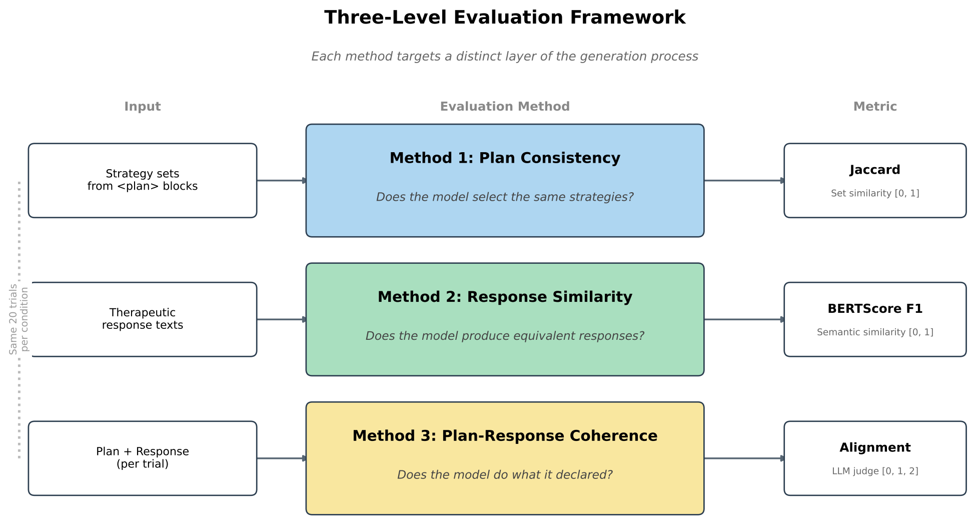

Figure 3.4.: Three-level evaluation framework. Each method targets a distinct layer of the generation process: strategy selection (Jaccard), response wording (BERTScore), and plan-response coherence (LLM judge).

The mean Jaccard score is computed over all $\binom{20}{2} = 190$ trial pairs per condition. Jaccard is appropriate because strategy combinations are unordered sets: a model that declares `confrontation / cognitive_reframe` has made the same clinical decision as one that declares `cognitive_reframe / confrontation`.

Because Jaccard operates on discrete sets of one to two items drawn from six categories, it takes only a small number of distinct values: $\{0, 0.33, 0.5, 0.67, 1.0\}$. This discreteness means Jaccard scores are ordinal rather than continuous, and descriptive statistics report medians and interquartile ranges rather than means and standard deviations.

Plan validity rate serves as an upstream quality gate. A malformed `<plan>` block that cannot be parsed into valid strategy identifiers is excluded from the Jaccard computation. The validity rate reports the proportion of trials with parseable plans, indicating whether the model can follow the structured output format at all.

3.7.2. Method 2: Response Similarity

Method 2 measures whether the model produces therapeutically equivalent responses across trials, independent of strategy declarations. Pairwise BERTScore F1 is computed over response texts (with plan blocks excluded) across the same 190 trial pairs.

The embedding model is DeBERTa-XLarge-MNLI (He et al., 2021). This model achieves the highest human correlation ($r = 0.778$) among more than 130 candidates on WMT16 segment-level evaluation. Its NLI fine-tuning aligns with the semantic similarity question: determining whether two therapeutic responses convey the same clinical content, regardless of surface phrasing. Its disentangled attention mechanism handles the surface-form variation typical of therapeutic language, where a competent response naturally varies in wording while maintaining the same clinical intent.

Surface metrics (BLEU, ROUGE) are not used. As established in section 2.4, token-overlap metrics penalise valid therapeutic paraphrasing and cannot distinguish semantically equivalent from semantically divergent responses when surface forms differ.

3.7.3. Method 3: Plan-Response Coherence

Method 3 measures whether the model's therapeutic response implements its declared strategy. A separate LLM judge scores each declared strategy on a ternary scale:

- **0 (absent):** The strategy is not reflected in the response.
- **1 (partial):** The strategy is mentioned or briefly addressed but not fully developed.
- **2 (implemented):** The strategy is clearly and substantively present in the response.

This ternary scale is borrowed from the clinical fidelity literature (section 2.1). The ENACT (Kohrt et al., 2015) and NIH BCC (Bellg et al., 2004) frameworks use the same three-level gradation (absent, partial, implemented) to assess whether a human therapist delivers an intervention correctly. Applying the same scale to LLM outputs maintains continuity with established clinical evaluation practice.

The judge model is Gemini 3.1 Pro, accessed through Google AI Studio at $T = 1.0$. Gemini is from a different model family than all evaluation targets, preventing self-preference bias. Its thinking mode supports structured reasoning about whether a response implements a given strategy. The judge receives the strategy definitions, the declared strategies, and the response text, and produces a score with explicit justification for each strategy. Gemini 3 Pro has been validated as a reliable judge for content validity assessment in psychology (Poźniak et al., 2026). The judge runs at $T = 1.0$, which is the Gemini 3 default. Google's developer documentation warns that lowering the temperature below 1.0 can cause looping or degraded output for these models (**W2RFM6E4**). Evaluation consistency is ensured through the structured ternary rubric and explicit scoring format.

Additional bias mitigations follow established recommendations (Zheng et al., 2023): cross-model judging ensures the judge never evaluates its own family, CoT justification requires the judge to explain its rating before assigning a score, and a transparent rubric anchors scores to specific behavioural criteria.

NLI cross-encoders were considered and rejected as an alternative. Preliminary testing on this task showed an F1 ceiling of approximately 0.55, insufficient for reliable scoring. Cross-encoders also produce no reasoning trace, making it impossible to audit individual judgements.

Method 3 is labelled exploratory. Judge reliability is itself an open question: the judge's ternary scores are not a ground truth but a computational proxy for clinical fidelity assessment. Validation proceeds through human annotation of a random subset of approximately 50 trials, scoring the same rubric. Cohen's κ between human and judge provides a reliability estimate. If κ is low, Method 3 results carry the caveat explicitly. In addition, 12 trials with the lowest coherence scores are manually reviewed to check whether the judge penalised valid therapeutic responses. A score of zero indicates a failed judge output rather than a genuine misalignment. In all observed cases, a single rerun of the judge call was sufficient to produce a valid score, so zero scores are treated as transient failures and rerun once before inclusion.

As reported in section 5.4, coherence scores are near ceiling across most models and temperature-independent. This means coherence serves primarily as a validation check (models consistently implement what they declare) rather than a comparative metric. This outcome validates Method 1: because plans are faithfully executed, Jaccard over plan sets measures genuine decision stability rather than label noise.

3.8. Model Selection

EU data sovereignty serves as the primary selection criterion for the study's main subject. The deployment context (reDreamAI) requires a model that can be self-hosted within EU data residency boundaries (Dale, 2025). All evaluation targets run in non-thinking mode for fair comparison.

The **primary subject** is Mistral Large 3 (675B Mixture of Experts (MoE), 41B active parameters, Apache 2.0). It is the EU sovereign flagship and the model that reDreamAI would deploy in production. The research question centres on whether this model achieves stability comparable to proprietary alternatives.

Two additional EU-sovereign models complete the Mistral comparison. Mistral Small 4 (119B MoE, 6.5B active parameters, Apache 2.0) is Mistral's newest release and the most efficient MoE in the set, activating fewer parameters per token than any other model. Mistral Small 3.2 (24B dense, Apache 2.0) serves as a predecessor baseline, showing whether the architectural shift from dense to MoE affects stability.

The **proprietary ceiling** comprises two models. GPT-5.4 (non-thinking variant) is the strongest proprietary model without thinking overhead. OpenAI reintroduced user-configurable temperature with the non-reasoning variants of GPT-5.2, making controlled temperature comparison possible. Claude Sonnet 4.6 adds a second proprietary reference point; Anthropic's character training emphasises thoughtful, safety-aware interaction, which may influence therapeutic stability independently of raw capability.

Open-weight comparators span three size classes and two architectures:

The selection provides size-class diversity (24B–675B), architectural diversity (dense vs. MoE), and provider diversity across five vendors. Qwen 3.5 and Mistral Small 4 are hybrid-thinking models that support optional reasoning. To maintain fair comparison, reasoning is disabled for all evaluation targets.

DeepSeek V3.2 was initially included as a large MoE comparator but was replaced by Qwen 3.5 397B. DeepSeek applies internal temperature scaling (API $T = 1.0$ maps to internal $T \approx 0.3$), which would have complicated cross-model temperature comparisons. Qwen 3.5 397B occupies the same architectural niche without this complexity, and including three Qwen scales (27B, 122B, 397B) enables within-family scaling analysis. Other models considered and dropped include Mistral Medium 3.1 (closed weights), GLM-5 (a third large MoE adds little signal), Nemotron 3 Super (requires $T = 1.0$ universally, complicating temperature comparison), and GPT-oss (mandatory reasoning mode cannot be disabled).

Table 3.2.: Evaluation targets. All models run in non-thinking mode via OpenRouter.

Group	Model	Size	Selection rationale
EU-sovereign (Mistral, Apache 2.0)			
Primary	Mistral Large 3	675B MoE (41B active)	EU-sovereign flagship. Production deployment candidate
EU	Mistral Small 4	119B MoE (6.5B active)	Newest Mistral. Fewest active parameters of any MoE in set
EU	Mistral Small 3.2	24B dense	Dense predecessor to Small 4
Open-weight comparators			
MoE	Qwen 3.5 122B	122B MoE (10B active)	Peer MoE comparator to Mistral Small 4
MoE	Qwen 3.5 397B	397B MoE (17B active)	Large MoE comparator to Mistral Large 3
Dense	Qwen 3.5 27B	27B dense	Same size class as Mistral Small 3.2
Dense	OLMo 3.1 32B	32B dense	Fully open (weights, data, code). Best provenance
Dense	Llama 3.3 70B	70B dense	Continuity with prior efficacy study
Proprietary ceiling			
Ceiling	Claude Sonnet 4.6	Proprietary	Character-trained for safety-aware interaction
Ceiling	GPT-5.4	Proprietary	Strongest non-thinking proprietary model

3.9. Experimental Conditions

The experiment follows a full factorial design:

Rather than testing only two temperature conditions, the experiment uses a five-point temperature sweep that spans from deterministic decoding to the vendor-recommended defaults. The scale is anchored at three points motivated by vendor documentation:

- $T = 0.0$ (greedy decoding): deterministic upper bound on stability. No reproducibility seed is set, as not all vendors support one. Even where a seed is available, GPU parallelisation and dynamic batching introduce floating-point non-determinism that makes output-level reproducibility best effort at most (Yuan et al., 2025). The twenty-trial repetition design captures this variance directly, following Blair-Stanek and Van Durme (2025), who use the same 20-trial protocol at $T = 0.0$ to quantify LLM stability.

- $T = 0.15$: recommended temperature for Mistral models. Mistral Small 3.2 sets $T = 0.15$ explicitly in its HuggingFace generation configuration. Mistral Large 3 and Mistral Small 4 lack a generation configuration default; their model cards recommend temperatures below 0.1 for production use, while code examples on the same cards use $T = 0.15$. Because no authoritative single default exists for these two models, 0.15 was chosen to match the verified Small 3.2 value and the vendor-provided code examples.

- $T = 0.6$: default temperature for Qwen 3.5 and OLMo 3.1, also close to the standard Llama chat range (0.6–0.7) and the proprietary defaults (GPT-5.4 and Sonnet 4.6 default to $T = 0.7$).

The intermediate points $T = 0.075$ and $T = 0.3$ fill the gap between Mistral's low default and the Qwen/Llama range. $T = 0.075$ was added after an initial four-point run revealed that several models peak in consistency between $T = 0.0$ and $T = 0.15$, suggesting that a small amount of sampling variance can be beneficial (discussed in section 5.2).

This five-point sweep replaces a conventional binary comparison ($T = 0.0$ vs. a single stochastic point). It reveals the full degradation curve rather than a single gap, and allows each model to be evaluated at and around its vendor-recommended operating temperature.

Table 3.3.: Experimental conditions. Total: $6 \times 5 \times 20 = 600$ trials per model, 6,000 across all ten models.

Dimension	Values
Vignettes	anxious, avoidant, cooperative, resistant, skeptic, trauma
Trials	20 per condition
Temperatures	$T \in \{0.0,\ 0.075,\ 0.15,\ 0.3,\ 0.6\}$
Models	10 evaluation targets (Table 3.2)

3.10. Statistical Analysis

The analysis is primarily descriptive. For each model and temperature condition, central tendency and dispersion are reported for all three evaluation metrics, aggregated across vignettes. Jaccard scores are ordinal (discrete values from a small set), so they are summarised with medians and interquartile ranges. BERTScore F1 and coherence scores are reported with means and standard deviations.

Spearman rank correlations serve two purposes. First, per-model correlations between Jaccard and temperature quantify the temperature sensitivity of each model across the five-point sweep. Second, cross-method correlations (Method 1 vs. Method 2, Method 1 vs. Method 3, Method 2 vs. Method 3) assess whether the three evaluation layers capture redundant or independent information. If they diverge, the clinical implications differ: high plan consistency with low response similarity indicates stable strategy selections but variable execution, while the reverse indicates different strategies producing similar surface outputs. Spearman rather than Pearson is used because Jaccard scores are ordinal and BERTScore distributions may be non-normal.

Kruskal-Wallis tests assess whether models and vignettes differ significantly on each metric. These are exploratory rather than confirmatory: the study is designed to describe stability patterns, not to test a preregistered hypothesis.

No fixed threshold defines “stable enough.” Clinical significance thresholds for LLM stability do not yet exist in the literature. Results are interpreted comparatively: model against model, across the temperature sweep. The random Jaccard baseline (expected value for random 2-of-6 selections, approximately 0.2) provides a lower reference point.

4. Implementation

This chapter describes how the methods from chapter 3 are realised in software. The pipeline is implemented in Python and structured around three concerns: a provider-agnostic LLM interface, a config-driven experiment runner, and a post-hoc aggregation pipeline. All source code is available in the project repository.

4.1. Software Architecture

The pipeline follows an async-first design. Every LLM call is non-blocking, and trials within a condition run concurrently via `asyncio.gather()`. This is a direct consequence of the factorial design: 600 trials per model would be prohibitive if executed sequentially. Parallel execution reduces wall-clock time from hours to minutes per model.

Configuration. Two YAML files define the full experiment declaratively. `models.yaml` specifies provider endpoints, model identifiers, role assignments, and evaluation targets. `experiment.yaml` controls sampling parameters (trial count, temperatures, maximum token length), metric selection, and output paths. Changing the evaluation target or temperature requires editing a single YAML field rather than modifying code. This separation ensures that experiment conditions are reproducible: the configuration snapshot saved with each run fully specifies how to recreate it.

Data validation. Pydantic models enforce type safety at every data boundary. The `Message` and `Conversation` classes validate all fields on construction and serialisation. When a frozen history is loaded from disk, `Conversation.from_dict()` uses `model_validate()` with `extra="ignore"`, silently discarding metadata fields that do not belong to the conversation schema. This is intentional: frozen histories may carry generation-time metadata (timestamps, model versions) that is irrelevant to evaluation.

Table 4.1.: Package structure. Each package maps to a distinct concern in the pipeline.

Package	Responsibility
`src/agents/`	Agent implementations (patient, therapist, router). Each agent extends a `BaseAgent` class that handles message formatting, system prompt injection, and provider delegation
`src/core/`	Data models (`Message`, `Conversation`), stage and language enums, YAML and prompt loaders
`src/llm/`	Provider abstraction. Single-class interface to all LLM backends
`src/stacks/`	Orchestration. `GenerationStack` runs the three-agent dialogue loop. `EvaluationStack` runs fused plan-and-response generation from frozen histories
`src/evaluation/`	Experiment runner, multi-trial sampler, and all three evaluation metrics (Jaccard, BERTScore, LLM judge coherence)

The same validation applies to plan extraction. The `extract_plan_strategies()` function parses the `<plan>` block from each trial output and validates each strategy identifier against the taxonomy. A malformed plan that cannot be parsed into valid strategy identifiers fails at extraction time rather than propagating downstream where it would silently corrupt Jaccard scores.

Provider abstraction. A single `LLMProvider` class wraps an `AsyncOpenAI` client and exposes one async method: `generate(messages, **kwargs)`, returning a tuple of response content and token usage. All ten evaluation targets, the patient agent, and the router use OpenRouter through this interface. The Gemini judge uses Google AI Studio through the same interface, since Google AI Studio exposes an OpenAI-compatible endpoint. Adding a new model or provider requires a configuration entry, not a code change.

A factory function (`create_provider()`) resolves role names (e. g. “therapist”, “judge”) to concrete provider configurations by looking up the active model assignment in `models.yaml`. For experiment-tier runs, the judge automatically upgrades from Gemini Flash Lite (used during development) to Gemini 3.1 Pro.

The codebase is organised into five packages:

4.2. Experiment Execution

An `ExperimentRun` object manages a single evaluation run: one model, one vignette, one temperature, twenty trials. On initialisation it creates a timestamped output directory and saves the configuration snapshot and frozen history.

Execution proceeds in three phases:

1. **Sampling.** A `Sampler` wraps the `EvaluationStack` and runs twenty independent trials. Each trial calls the therapist model once (fused mode), producing a `<plan>` block and a therapeutic response. Trials run in parallel via `asyncio.gather()` with built-in retry logic for rate-limit errors (HTTP 429) using exponential backoff.

2. **Metric computation.** After all trials complete, the run computes all three evaluation metrics. Method 1 (Jaccard) and Method 2 (BERTScore) are computed locally. Method 3 (coherence) issues parallel async calls to the GEMINI judge, with a 120-second timeout per trial to handle unresponsive judge calls gracefully.

3. **Persistence.** Results are saved to the run directory.

Each run produces a self-contained output bundle:

```
experiments/runs/{timestamp}_{model}_{vignette}/
    config.yaml              # run parameters (model, temp, slice)
    frozen_history.json      # input conversation (irreplaceable)
    trials/
        trial_01.json        # plan + response + token usage
        ...
        trial_20.json
    metrics.json             # Jaccard, BERTScore, validity, coherence
    judgments.json            # raw judge outputs with reasoning
```

The frozen history is the only irreplaceable artefact in this structure. If the evaluation logic changes (a bug fix in Jaccard computation, a new judge prompt), metrics and judgements can be recomputed from the saved trials. Trials themselves can be recomputed from the frozen history plus the configuration, though at the cost of additional Application Programming Interface (API) calls.

The plan extraction logic handles several output formats that different models produce. The primary pattern matches a closed `<plan>` tag. Fallback patterns handle unclosed tags, bare tag-style strategy names, and plain-text first-line declarations. This robustness is necessary because the ten evaluation targets do not all follow the structured output format identically, despite receiving the same system prompt.

4.3. Aggregation Pipeline

After all experiment runs complete, an aggregation script collects results into a single analysis frame. The script iterates over all run directories, loads each `config.yaml` and `metrics.json`, and produces one row per run indexed by model, vignette, slice, and temperature.

Strategy counts are expanded into six separate columns (one per taxonomy category), enabling per-strategy frequency analysis. Alignment scores per strategy are similarly expanded. The resulting DataFrame is saved as a CSV file (`stats/data/all_runs.csv`) that serves as the input for all downstream analysis and visualisation scripts.

This unified frame is a prerequisite for the cross-method analysis in section 5.5. Spearman correlations between Method 1, Method 2, and Method 3 require all three metrics aligned on the same condition index. Without a single frame that maps each condition to its Jaccard, BERTScore, and alignment scores, cross-method comparison would require manual alignment across separate output files.

Downstream scripts in `stats/scripts/` consume this CSV to produce descriptive statistics (`descriptives.py`), statistical tests (`tests.py`), and all five result figures. The separation between aggregation and analysis allows each step to be rerun independently: if a figure script is modified, only visualisation reruns, not the full aggregation.

5. Results

All results are reported at the second rescripting-turn boundary (`slice_2`), aggregated across six vignettes. Each data point represents the mean of twenty independent trials. Full per-condition tables appear in Appendix C.

5.1. Plan Validity and Strategy Distribution

Before analysing stability, the plan extraction step must be validated. A trial is valid if the model produces a well-formed plan block containing at least one recognised strategy category. Across all 6,000 trials, validity rates exceed 99% for every model. The lowest rates belong to Qwen 3.5 27B (99.3%) and OLMo 3.1 32B (99.7%). All other models achieve 99.8% or above. Validity does not vary meaningfully with temperature. These rates confirm that the fused chain-of-thought prompt reliably elicits structured plan output, and that the subsequent stability metrics are computed over near-complete data.

Strategy distributions vary across models (Figure 5.1). All ten models draw from the same six-category taxonomy, but the relative frequencies differ. `Confrontation` and `self_empowerment` dominate across most models. GPT-5.4 and Sonnet 4.6 show the narrowest distributions, concentrating on two to three categories. Qwen 3.5 27B and Llama 3.3 70B spread selections more evenly across categories. These distributional differences are a prerequisite for interpreting the consistency results that follow: a model that always selects the same two strategies will score high on Jaccard regardless of temperature, while a model that draws from a wider repertoire has more room to vary.

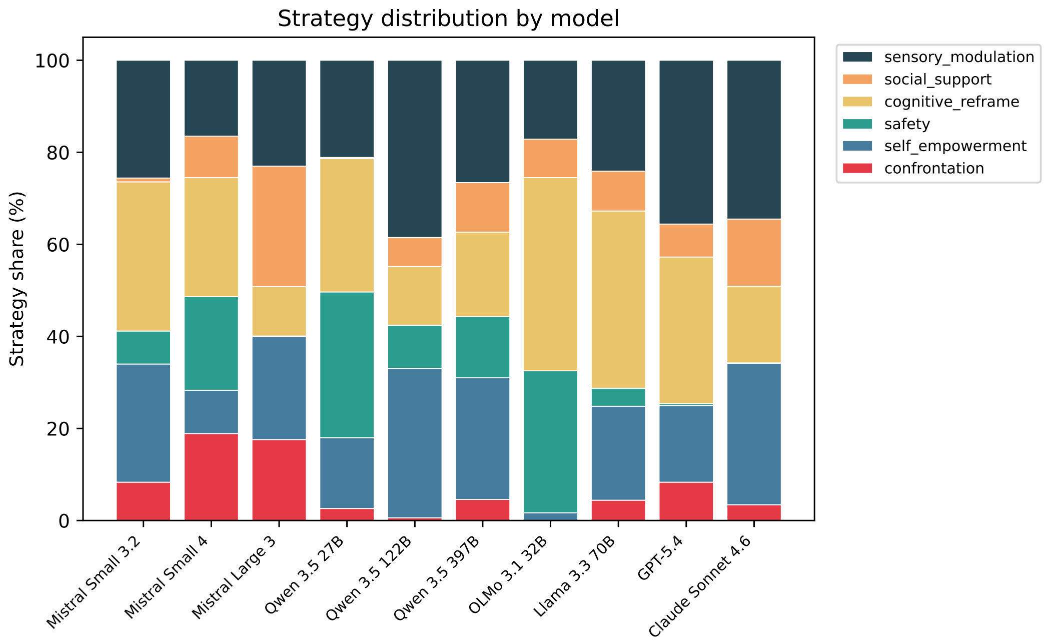

Figure 5.1.: Strategy distribution by model, pooled across temperatures and vignettes. Each bar shows the share of trials selecting each taxonomy category. Models with narrower distributions have less room to vary.

5.2. Plan Consistency (Method 1)

Figure 5.2 presents the headline result: mean Jaccard similarity by model and temperature, grouped by model family.

Figure 5.3 complements the Jaccard measure with modal-set agreement: the proportion of trials that select the most common strategy combination. Where Jaccard measures average pairwise overlap, modal agreement measures how often the plurality strategy dominates.

Proprietary models are temperature-immune. GPT-5.4 and Sonnet 4.6 achieve $J = 1.000$ at every temperature point. Sampling temperature does not affect their strategy selections at all.

Most open-weight models peak at low temperatures. The Mistral family shows a characteristic pattern: Mistral Small 3.2 maintains $J = 1.000$ from $T = 0.0$ through $T = 0.3$ and drops only at $T = 0.6$ ($J = 0.933$). Mistral Large 3 scores $J = 0.933$ at $T = 0.0$, rises to $J = 1.000$ at $T = 0.075$ and $T = 0.15$, then declines

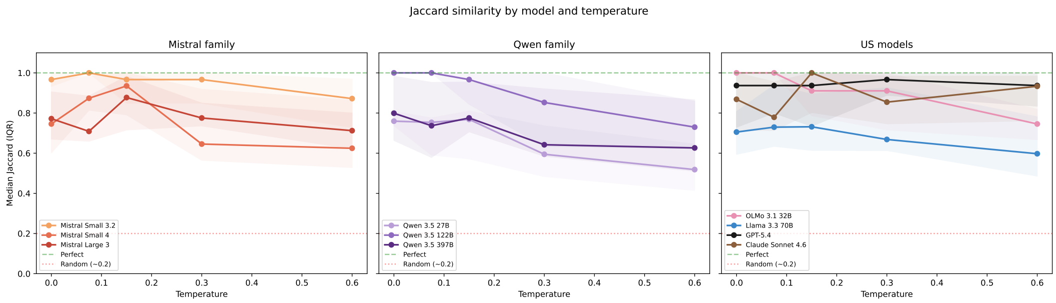

Figure 5.2.: Mean Jaccard similarity by model and temperature, grouped by family. Shaded bands show ±1 SD across vignettes. The dashed line marks the random baseline ($J \approx 0.2$). Higher is more consistent.

to $J = 0.652$ at $T = 0.6$. Mistral Small 4 shows the most pronounced version of this pattern: $J = 0.722$ at $T = 0.0$, jumping to $J = 1.000$ at $T = 0.075$ before declining.

Greedy decoding is not always optimal. Three models score higher at $T = 0.075$ than at $T = 0.0$: Mistral Small 4 ($0.722 \to 1.000$), Mistral Large 3 ($0.933 \to 1.000$), and Llama 3.3 70B ($0.756 \to 0.881$). A small amount of sampling variance produces more consistent strategy selection than pure greedy decoding for these models.

Temperature sensitivity varies by model. Per-model Spearman correlations between Jaccard and temperature (Table 5.1) reveal three groups. OLMo 3.1 32B ($\rho = -0.632$, $p < 0.001$), Qwen 3.5 27B ($\rho = -0.576$, $p = 0.001$), and Qwen 3.5 122B ($\rho = -0.513$, $p = 0.009$) show statistically significant degradation. All other models, including the entire Mistral family, show no significant temperature effect on Jaccard. The pooled correlation across all models is $\rho = -0.228$ ($p < 0.001$).

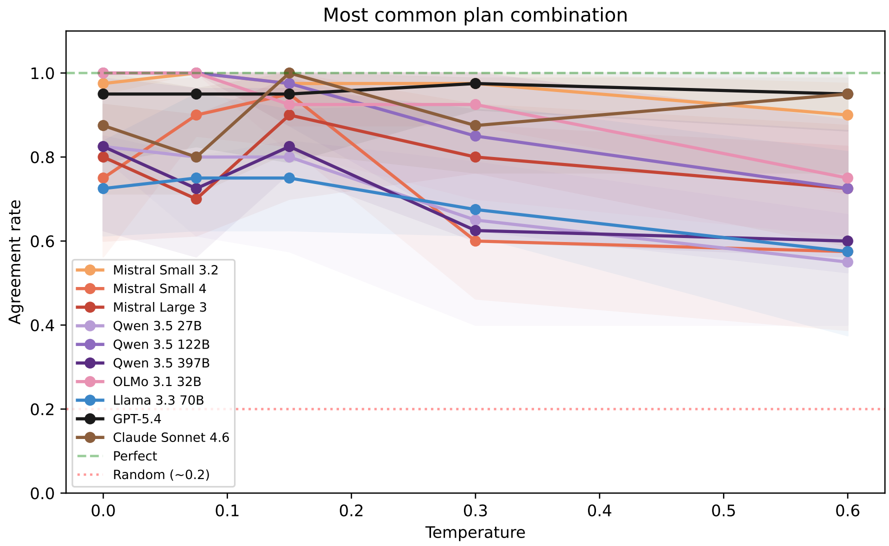

Figure 5.3.: Modal-set agreement by model and temperature. The agreement rate is the proportion of trials selecting the most common strategy combination per condition. Models with agreement near 1.0 select the same strategy pair on nearly every trial.

Table 5.1.: Spearman correlation between Jaccard and temperature, per model. Significant correlations ($p < 0.05$) are marked with an asterisk.

Model	ρ	p	Sig.
Mistral Small 3.2	−0.190	0.315	
Mistral Small 4	−0.196	0.299	
Mistral Large 3	−0.277	0.139	
Qwen 3.5 27B	−0.576	0.001	*
Qwen 3.5 122B	−0.513	0.009	*
Qwen 3.5 397B	−0.232	0.217	
OLMo 3.1 32B	−0.632	<0.001	*
Llama 3.3 70B	−0.153	0.421	
GPT-5.4	+0.075	0.695	
Claude Sonnet 4.6	−0.063	0.741	
Pooled	−0.228	<0.001	*

5.3. Response Similarity (Method 2)

BERTScore F1 measures semantic similarity between response texts, independent of the plan block. The pooled temperature effect is stronger than for Jaccard ($\rho = -0.486$, $p < 0.001$): rising temperature changes response wording more than it changes strategy selection.

However, BERTScore has limited sensitivity to plan-level differences. The slice depth analysis (section 5.6) confirms this: across conversation depths, Jaccard varies visibly while BERTScore remains flat. A model that switches from `confrontation` to `cognitive_reframe` changes its therapeutic plan but produces surface text similar enough that BERTScore does not detect the difference.

BERTScore therefore captures response *style* consistency rather than decision *content* consistency. It serves as a sanity check: if BERTScore were low, the model would be generating erratic text regardless of strategy. The substantive signal for therapeutic stability comes from Jaccard and coherence.

Full BERTScore results by model and temperature appear in Appendix C.

5.4. Plan-Response Coherence (Method 3)

Figure 5.4 shows plan-response coherence by temperature and per model. Coherence measures whether the model's response implements the strategies it declared in its plan.

The central finding is that coherence is temperature-independent. The pooled Spearman correlation between coherence and temperature is $\rho = -0.053$ ($p = 0.361$), not significant. Models do not get worse at implementing their declared strategies as temperature rises. They select *different* strategies more often (lower Jaccard) but implement whichever strategies they select equally well (stable coherence).

This separation has a clinical implication: temperature-induced instability occurs at the decision-making level, not at the execution level. A model at high temperature does not produce incoherent responses. It produces coherent responses to a different therapeutic plan.

Kruskal-Wallis tests confirm that models differ significantly on coherence ($p < 0.001$). Mistral Small 4 shows the lowest mean coherence, consistent with its erratic strategy selection at extreme temperatures. The proprietary models and Mistral Small 3.2 cluster near the top.

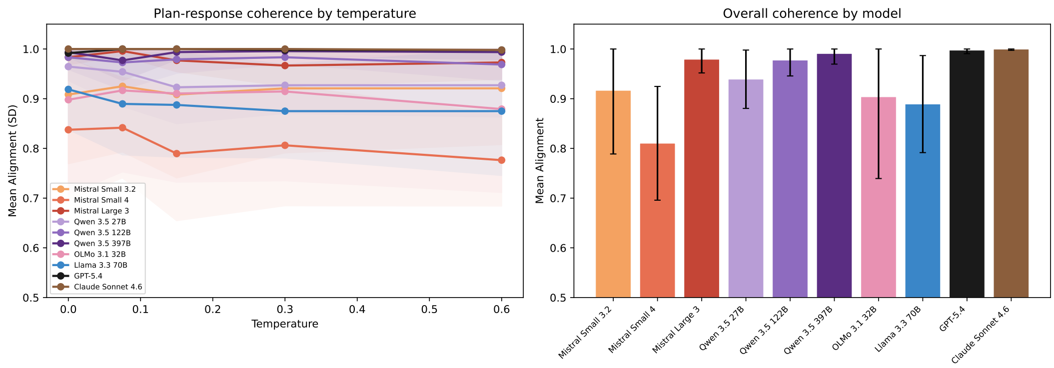

Figure 5.4.: Plan-response coherence by temperature (left) and overall by model (right). Error bars show ± 1 SD. Most models implement their declared strategies consistently regardless of temperature.

5.5. Cross-Method Analysis

The three evaluation methods are related but not redundant. Table 5.2 reports Spearman correlations between all method pairs, and Figure 5.5 visualises the relationship between plan consistency and response similarity.

The moderate Jaccard–BERTScore correlation ($\rho = 0.560$) indicates that models with more consistent strategy selection also tend to produce more similar response text, but the relationship is not strong enough for one metric to substitute for the other. The weak Jaccard–Coherence correlation ($\rho = 0.285$) confirms that plan consistency and plan-response coherence measure different properties.

The scatter plot (Figure 5.5) reveals the model landscape. GPT-5.4 and Sonnet 4.6 anchor the upper-right corner: high on both dimensions. Qwen 3.5 27B sits in the lower-left: low plan consistency and low response similarity. Mistral Small 4 presents an interesting case: moderate Jaccard but the lowest BERTScore in the set, suggesting that even when it selects the same strategies, its response wording varies considerably.

Table 5.2.: Spearman correlations between the three evaluation methods. All correlations are significant ($p < 0.001$). The moderate Jaccard–BERTScore correlation and weak Jaccard–Coherence correlation confirm that the methods capture distinct dimensions.

Method pair	ρ	Interpretation
Jaccard vs. BERTScore	0.560	Moderate
Jaccard vs. Coherence	0.285	Weak
BERTScore vs. Coherence	0.376	Weak–moderate

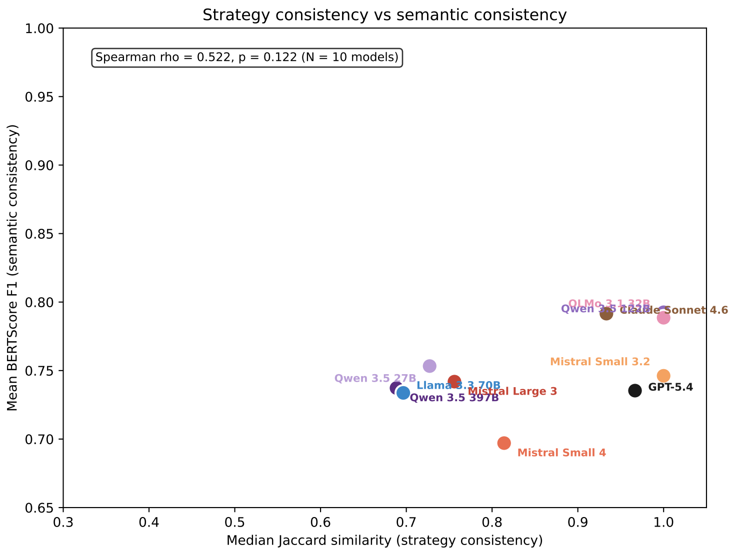

Figure 5.5.: Plan consistency (median Jaccard) versus response similarity (mean BERTScore F1) per model, pooled across temperatures. The proprietary models cluster in the upper-right corner. Qwen 3.5 27B and Mistral Small 4 occupy opposite ends of the BERTScore axis despite similar Jaccard ranges.

5.6. Secondary Effects

5.6.1. Vignette Effects

Kruskal-Wallis tests detect a significant vignette effect on Jaccard ($p = 0.007$) and BERTScore ($p = 0.015$). Some patient profiles produce more consistent therapeutic responses than others across models. Full per-vignette heatmaps appear in Appendix C.

5.6.2. Conversation Depth

A preliminary analysis tested Mistral Large 3 at five conversation depths (slices 1–5) across all six vignettes at $T = 0.075$ and $T = 0.15$. Figure 5.6 shows the results.

Jaccard consistency is lowest at the second slice (mean $J = 0.77$ at $T = 0.075$) and rises to near-ceiling values from the third slice onward ($J > 0.93$). BERTScore and coherence remain flat across all depths, confirming that conversation depth affects strategy *selection* but not response *wording* or plan *execution*.

The main experiment evaluates at slice 2, making the reported stability scores a conservative estimate. A model that achieves high consistency at this depth would likely score even higher at later conversation points where prior exchanges provide more anchoring context.

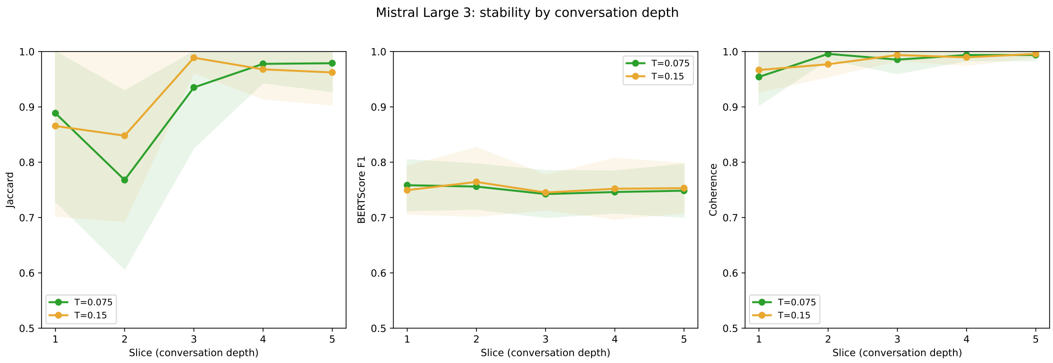

Figure 5.6.: Stability by conversation depth for MISTRAL LARGE 3. Jaccard is lowest at slice 2 and rises to near-ceiling values from slice 3 onward. BERTScore and coherence remain flat across depths.

6. Discussion

6.1. Interpreting the Three-Level Framework

The three evaluation methods were designed to capture distinct aspects of therapeutic stability (section 3.7). The results confirm that they do: the cross-method correlations are moderate at best ($\rho = 0.560$ for Jaccard–BERTScore, $\rho = 0.285$ for Jaccard–Coherence). Had any two methods been strongly correlated, one would be redundant. Instead, each method reveals information that the others miss.

The most informative finding is the separation between plan consistency and coherence. Temperature increases reduce Jaccard (models select different strategies) but do not reduce coherence (models implement whichever strategy they select equally well). This pattern localises the source of instability to the *decision-making* layer rather than the *execution* layer. A model at high temperature does not produce incoherent or unfaithful responses. It produces coherent responses to a different therapeutic plan.

This distinction matters in practice. If the model selected consistent strategies but failed to carry them out, the problem would be in the response generation. Instead, the pattern observed here is the opposite: the model carries out its plans faithfully but picks different plans across runs. The instability sits in strategy selection, not in response quality. Knowing this points toward temperature tuning rather than prompt redesign as the more promising adjustment.

BERTScore occupies a more limited role than initially anticipated. It correlates moderately with Jaccard but cannot distinguish between clinically different plans when the surface text remains similar. The slice depth analysis confirms this: Jaccard varies across conversation depths while BERTScore stays flat (Figure 5.6). A model that switches from `confrontation` to `cognitive_reframe` changes its therapeutic approach, but the response text remains similar enough that BERTScore does not detect the difference. BERTScore therefore serves primarily as a sanity check for response coherence rather than as an independent stability measure.

6.2. Cross-Model Comparison

The central question of this thesis is whether the EU-sovereign flagship, Mistral Large 3, achieves therapeutic stability comparable to proprietary alternatives (Figure 5.2).

At the vendor-recommended temperature ($T = 0.15$), Mistral Large 3 achieves $J = 1.000$, matching GPT-5.4 and Sonnet 4.6. At its peak ($T = 0.075$), it also reaches $J = 1.000$. The sovereignty tradeoff, at low temperatures, is negligible: the EU flagship matches the proprietary ceiling on plan consistency.

The tradeoff becomes visible at higher temperatures. At $T = 0.6$, Mistral Large 3 drops to $J = 0.652$, while the proprietary models maintain $J = 1.000$. However, $T = 0.6$ is well above Mistral's recommended operating range. A deployment that follows vendor guidance ($T \leq 0.15$) would not encounter this degradation.

Mistral Small 3.2 (24B dense) is the most stable open-weight model in the set, maintaining $J = 1.000$ through $T = 0.3$ and showing no statistically significant temperature sensitivity.

Mistral Small 4 presents the most pronounced version of the greedy-suboptimality pattern. At $T = 0.0$ it scores only $J = 0.722$, then jumps to $J = 1.000$ at $T = 0.075$. A small amount of sampling variance helps the model settle on a consistent strategy rather than oscillating between candidates under greedy decoding. Mistral Large 3 ($0.933 \rightarrow 1.000$) and Llama 3.3 70B ($0.756 \rightarrow 0.881$) show the same direction.

Among the open-weight comparators, parameter count does not straightforwardly predict stability. Qwen 3.5 122B (10B active) outperforms Qwen 3.5 397B (17B active) at every temperature point. The two models differ in how much of their total capacity is used per token (sparsity ratio): the 122B model activates 8.2% of its parameters, while the 397B model activates only 4.3% (routing through 512 experts). A lower activation ratio means more of the output depends on which experts the router selects, which may make the model more sensitive to sampling randomness. This remains a hypothesis; confirming it would require analysis at the expert routing level. OLMo 3.1 32B matches larger models at low temperatures but degrades significantly at higher ones ($\rho = -0.632$, strongest temperature sensitivity in the set). The Qwen 3.5 family shows a consistent pattern across its three size classes: the mid-size 122B model is the most stable, while the smallest (27B) and largest (397B) are less consistent.

The proprietary models (GPT-5.4, Sonnet 4.6) form a distinct tier. Their perfect Jaccard across all temperatures suggests that character training, safety-layer alignment, or other proprietary techniques create a stability floor that sampling temperature cannot disrupt. Whether this reflects genuine robustness or a narrower output distribution (fewer strategies available to vary) cannot be determined from these results alone.

6.3. Temperature and Clinical Deployment

The five-point temperature sweep reveals that the relationship between temperature and stability is not monotonic for all models. Three models achieve their peak Jaccard at $T = 0.075$ rather than $T = 0.0$, meaning greedy decoding is actively counterproductive for them: Mistral Small 4 ($0.722 \rightarrow 1.000$), Mistral Large 3 ($0.933 \rightarrow 1.000$), and Llama 3.3 70B ($0.756 \rightarrow 0.881$). For these models, defaulting to $T = 0.0$ for maximum stability may be counterproductive. This is consistent with Mistral's optimisation for low non-zero temperatures: the vendor default sits at $T = 0.15$, and the results suggest that stability-sensitive tasks are best served in the range between zero and that default.

The pooled Spearman correlation ($\rho = -0.228$, $p < 0.001$, Table 5.1) confirms that higher temperature generally reduces plan consistency, but the effect is weak. Per-model analysis reveals that only three models show statistically significant degradation: OLMo 3.1 32B, Qwen 3.5 27B, and Qwen 3.5 122B. The remaining seven models, including the entire Mistral family, show no significant temperature effect on Jaccard.

For clinical deployment, these results suggest a practical recommendation: operate at or near the vendor-recommended temperature. Models designed for a specific temperature range tend to be most stable within it. Pushing a model to $T = 0.0$ in the hope of maximising determinism can be counterproductive.

Coherence is temperature-independent across all models ($\rho = -0.053$, $p = 0.361$). This means that the clinical cost of higher temperature is limited to strategy *selection* variability. The model does not become worse at *executing* its chosen strategy. A patient receiving care from a high-temperature model would encounter different therapeutic approaches across sessions, but each individual session would be internally coherent.

6.4. Vignette-Dependent Behaviour

The significant vignette effect on Jaccard ($p = 0.007$) and BERTScore ($p = 0.015$) indicates that some patient profiles are harder to treat consistently than others. This is clinically relevant: if a model is stable for cooperative patients but variable for resistant or trauma-presenting patients, deployment safety depends on the patient population being served.

This finding suggests that stability evaluation should not rely solely on aggregate metrics. Per-vignette reporting, as provided in this study's appendix, is necessary to identify the conditions under which a model's consistency breaks down. A model that appears stable on average may be unreliable for the specific patient profiles that need the most careful treatment.

Full per-vignette analysis is provided in Appendix C.

6.5. Evaluation Framework Reflections

Strengths. Measuring at three layers revealed that strategy selection varies while execution stays faithful. BERTScore alone would have missed this, since it captures therapeutic tone but not the underlying strategy choices (section 5.3). Frozen histories and 20-trial repetition (Blair-Stanek & Van Durme, 2025) ensure that observed variance comes from the model, not from differences in input or sample size.

Taxonomy sensitivity. Consistency scores depend on how strategies are categorised. Broader categories inflate Jaccard (more selections land in the same bin), finer categories deflate it. The six-category taxonomy (Table 3.1) balances clinical detail with the ability to tell strategies apart, but different category boundaries would shift the absolute numbers. The relative ordering of models is more stable, since all are evaluated against the same set.

Declared intent, not cognition. The plan block captures what the model declares, not what it internally computes (section 3.5). It cannot tell apart a model that picks a strategy and then writes the response from one that writes a response and then picks a matching label. The high coherence scores fit either reading. What matters here is that declarations vary across runs, and that variation predicts variable therapeutic behaviour regardless of the underlying process.

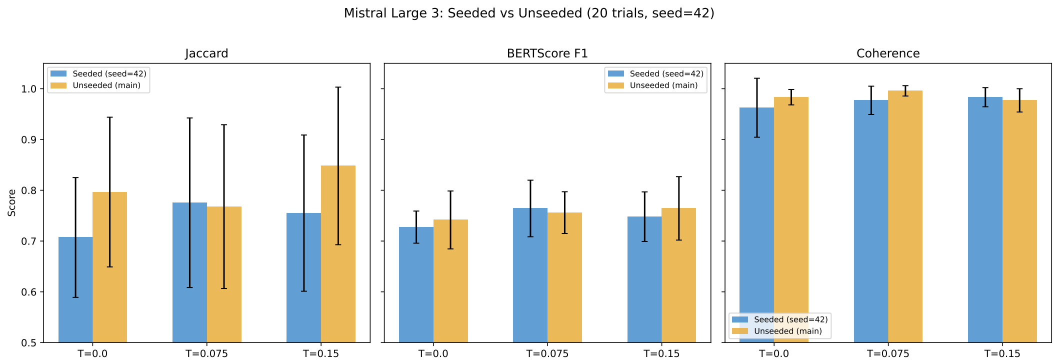

Figure 6.1.: Seed comparison for MISTRAL LARGE 3. Setting a fixed seed (`seed=42`) does not improve Jaccard, BERTScore, or coherence compared to unseeded runs. Error bars show ±1 SD across vignettes. Differences are within noise.

Generalisability. The framework is not specific to IRT. Other structured therapy protocols where the therapist chooses from a defined set of techniques could be evaluated the same way. CBT and motivational interviewing are likely candidates. Each would need its own strategy taxonomy, but the evaluation setup and metrics carry over.

Ethical scope. This framework measures whether a model produces *consistent* therapeutic responses, not whether those responses are *clinically appropriate*. A model that consistently selects the wrong strategy scores high on plan consistency. Stability is a necessary precondition for clinical deployment, as stated in section 1.2, but it is not sufficient. All evaluations use synthetic patients and no human participants are involved. Clinical validation through human expert evaluation remains necessary before any deployment to real patients.

Methodological constraints. No reproducibility seed was set, as not all vendors support one. Even where seeds are available, they provide best-effort determinism at most (Yuan et al., 2025). A supplementary seed experiment on MISTRAL LARGE 3 confirms this: setting a fixed seed (`seed=42`) does not improve consistency compared to unseeded runs (Figure 6.1). The trial repetition design captures stochastic variance directly.

The experiment evaluates at a single conversation depth (`slice_2`), chosen as a conservative estimate based on the preliminary depth analysis (section 3.3). Results at other depths may differ. All models were accessed through OpenRouter, so provider-side batching and load balancing may contribute to the non-determinism observed at $T = 0.0$, though this mirrors realistic deployment conditions.

Further limitations regarding scope, vignette coverage, judge validation, and taxonomy constraints are discussed in section 7.2.

7. Conclusion

7.1. Summary of Findings

This thesis asked how consistent LLM therapeutic responses are across independent runs within a structured IRT protocol. Three evaluation methods were applied to ten models across 6,000 trials spanning five temperature conditions.

The main findings are:

1. **Consistency varies widely across models and temperatures.** Some models achieve perfect plan consistency ($J = 1.000$) at all temperatures, while others drop below $J = 0.5$ at higher settings. Temperature is the primary driver of variation for open-weight models.

2. **The EU-sovereign flagship matches proprietary models at recommended temperatures.** Mistral Large 3 achieves $J = 1.000$ at $T = 0.075$ and $T = 0.15$, equal to GPT-5.4 and Sonnet 4.6. The gap only appears above the recommended operating range.

3. **Instability sits in strategy selection, not execution.** Temperature reduces plan consistency but not plan-response coherence. Models pick different strategies at higher temperatures but carry out whichever strategy they pick equally well.

4. **Greedy decoding is not always optimal.** Three models achieve higher consistency at $T = 0.075$ than at $T = 0.0$. Whether this reflects a genuine benefit of low sampling variance or an artefact of the 20-trial sample size cannot be determined from these results alone.

5. **No single metric captures the full picture.** The three evaluation levels are only moderately correlated ($\rho = 0.285$ to 0.560). BERTScore misses plan-level changes.

6. **Patient profile matters.** A significant vignette effect ($p = 0.007$) means that stability evaluations based on aggregate metrics alone may miss per-profile weaknesses.

Regarding the two contributions stated in section 1.4: the three-level framework (Contribution 1) fulfilled its design goal by identifying that instability is concentrated at the strategy selection layer, a finding that neither BERTScore nor coherence alone could have revealed. The cross-model comparison (Contribution 2) demonstrates that the EU-sovereign Mistral Large 3 achieves stability comparable to proprietary models at its recommended operating temperature, supporting its viability for clinical deployment under sovereignty constraints.

7.2. Limitations

The study evaluates a single phase (rescripting) of a single protocol (IRT). The rescripting phase was selected because it requires active strategy decisions, but stability in other stages or other therapeutic protocols may differ.

All patient interactions are simulated. Synthetic patients follow scripted behavioural profiles and do not exhibit the unpredictable responses of real patients. Six profiles sample the difficulty range (section 3.2) but cannot claim exhaustive coverage. No human participants were involved in this study, and the results measure computational consistency rather than clinical safety or efficacy. IRT targets nightmares rather than psychiatric diagnoses, which limits clinical risk, but any deployment to real patients would require validation beyond what this framework provides.

The LLM judge (Gemini 3.1 Pro) has only been validated on a small subset of trials due to the volume of data (6,000 trials). Method 3 results should be interpreted as a computational estimate rather than a clinically validated measure.

The six-category strategy taxonomy constrains what can be measured. Strategies that fall outside these categories are invisible to Method 1, and different category boundaries would produce different absolute consistency scores.

Finally, consistency is not the same as quality. A model that consistently selects an inappropriate strategy scores high on plan consistency. Stability is necessary for clinical deployment but not sufficient on its own.

7.3. Future Work

Several directions follow from this work.

Extending the evaluation beyond the rescripting phase to all IRT stages would test whether the observed patterns hold when the model faces different types of clinical decisions. More broadly, the three-level framework applies to other therapy protocols where the therapist chooses from a defined set of techniques, such as CBT or motivational interviewing. Each would need its own strategy taxonomy, but the evaluation setup carries over.

Since model vendors update weights and infrastructure without notice, repeating the evaluation across model versions would show whether stability improves or fluctuates over time. reDreamAI targets German-speaking users, so evaluating stability in German would test whether the English-language patterns transfer.

On the measurement side, a larger human annotation study against the same ternary rubric would strengthen the judge validation that this study could only perform on a small subset. Comparing human and LLM judge scores would clarify how much trust to place in automated coherence assessment.

Finally, newer model architectures may affect output consistency independently of temperature. Qwen 3.5 already uses Gated DeltaNet (Yang et al., 2025), a hybrid attention mechanism where gated memory updates control how much prior context is retained or forgotten. In principle, this could reduce the effect where an early token divergence propagates through the rest of the response. The results here do not show a clear stability advantage for Qwen 3.5 over standard transformers, but whether architectural choices can improve therapeutic consistency beyond what temperature tuning achieves remains open.

Acknowledgements

Use of AI Tools in This Thesis

Throughout the research and writing of this thesis, I made use of several LLM-based tools in a supporting capacity.

For intellectual engagement with the subject matter, I used Claude (Anthropic)[1] and ChatGPT (OpenAI)[2] as discussion partners. Their role was to challenge my reasoning, surface counterarguments, and suggest alternative framings for the arguments I had already developed. As English is not my native language, I occasionally used these tools to refine phrasing where my original formulations lacked precision or fluency.

For software development, I used Claude Code and Cursor (with Claude Sonnet) as coding assistants for the evaluation pipeline that forms the core of this thesis. Given that the pipeline itself measures LLM behavioral stability, I took particular care to verify all AI-generated code before inclusion, to avoid introducing confounds into the instrument being studied.

No part of the thesis text was generated by AI. All arguments, interpretations, and conclusions are my own.

[1] https://claude.ai
[2] https://chatgpt.com

Glossary

API Application Programming Interface

BDI Brief Dreamer Interview

CBT Cognitive Behavioral Therapy

CoT Chain-of-Thought

EU European Union

GDPR General Data Protection Regulation

IRT Imagery Rehearsal Therapy

JSON JavaScript Object Notation

LLM Large Language Model

MoE Mixture of Experts

NLI Natural Language Inference

NLP Natural Language Processing

PTSD Post-Traumatic Stress Disorder

RLHF Reinforcement Learning with Human Feedback

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A. Strategy Taxonomy

The final six-category taxonomy (v4) constrains the `<plan>` block to the following therapeutic rescripting mechanisms. Each category describes *how* the nightmare changes, not what outcome it achieves. The taxonomy evolved through four iterations to resolve distribution skew (section 3.6).

Table A.1.: Strategy taxonomy v4: full definitions as provided to the therapist model. NOT-examples reduce category overlap.

Category	Definition (as given to the model)
`confrontation`	The dreamer directly faces, fights, or stands up to the threat itself (NOT avoiding, escaping, or gaining general confidence).
`self_empowerment`	The dreamer gains a new ability, skill, or inner strength they did not have before (e. g. discovers they can fly, becomes stronger, unlocks a hidden power). The DREAMER changes, not the threat. (NOT reinterpreting existing elements, which is `cognitive_reframe`.)
`safety`	A physical barrier, shelter, or protective figure is added to shield the dreamer (e. g. walls, locked doors, a guardian). NOT just feeling calm or changing the scene.
`cognitive_reframe`	The meaning or identity of existing dream elements changes (e. g. a monster becomes harmless, an enemy becomes a friend). The THREAT changes, not the dreamer. (NOT the dreamer gaining new powers, which is `self_empowerment`.)
`social_support`	People in the dream become supportive, or a new ally is introduced. Existing characters can shift from threatening to encouraging. Any form of interpersonal support counts.
`sensory_modulation`	The dream's sensory qualities change: what is seen, heard, or felt shifts (e. g. darkness becomes light, harsh sounds become soft music, the dreamer feels calm through breathing or self-talk). Covers both external environment changes and internal somatic calming.

Revision history.

v1 (8 categories): Included `empowerment` and `mastery` as separate categories. These two consumed 87.7% of all strategy picks across 1,272 trials, while three categories were never selected. Root cause: both describe a therapeutic *goal* (the patient gains control) rather than a *mechanism*.

v2 (7 categories): Merged `empowerment` and `mastery` into `agency`. The merged category achieved 100% dominance.

v3 (7 categories): Split `agency` along the mechanism dimension into `confrontation` (external action) and `self_empowerment` (internal transformation). Sharpened all descriptions to focus on mechanism, not outcome.

v4 (6 categories): Merged `emotional_regulation` into `sensory_modulation`. In smoke testing (120 trials), `emotional_regulation` appeared once (0.4%). In single-turn rescripting, internal calming manifests as sensory change.

B. Prompts

This appendix reproduces the key prompts used in the evaluation pipeline. All prompts are stored as YAML files in `data/prompts/`.

B.1. Fused Plan and Response Prompt

The following system prompt is injected into the therapist model during evaluation. The `{categories_block}` placeholder is replaced with the full taxonomy definitions from Table A.1.

```
You are an Imagery Rehearsal Therapy (IRT) assistant
helping a patient rescript their nightmare.

First, select 1-2 therapeutic strategies from this taxonomy:
{categories_block}

Choose the strategies most specifically suited to THIS
patient's nightmare content. Avoid defaulting to general
approaches when a more specific mechanism fits.

Then respond to the patient using those strategies.

STRICT output format:
Line 1: <plan>category1 / category2</plan>
Line 2: (empty)
Line 3+: Your therapeutic response (conversational,
         supportive, max 3 sentences)
```

B.2. Coherence Judge Prompt

The following prompt is sent to Gemini 3.1 Pro ($T = 1.0$, thinking enabled). The `{taxonomy_block}` is replaced with the full strategy definitions. The `{strategies_block}` and `{response}` are filled per trial.

```
System:
You are evaluating whether a therapist's response implements
specific therapeutic strategies from IRT.

## Strategy Definitions
{taxonomy_block}

## Scoring Rubric
For each declared strategy, assign ONE score:
- 2 (implemented): The response clearly and specifically
  implements this strategy.
- 1 (partial): The response touches on this strategy but
  does not fully develop it.
- 0 (absent): The response does not implement this strategy.

## Instructions
1. Read the therapist response carefully.
2. For each declared strategy, provide brief reasoning
   (1-2 sentences), then your score.
3. Use EXACTLY this output format for each strategy:

strategy_id: <reasoning> | score: <0|1|2>

User:
## Declared Strategies
{strategies_block}

## Therapist Response
{response}
```

B.3. Example Patient Vignette

Six vignettes define patient profiles for dialogue generation. The following example shows the *anxious* profile. All vignettes follow the same schema.


```
{
  "type": "anxious",
  "name": "Maya",
  "age": 24,
  "gender": "female",
  "background": "A graduate student dealing with nightmares
```

```
    that started during a particularly stressful thesis
    period. Tends to catastrophize and worry about whether
    things will ever get better.",
  "nightmare": {
    "content": "Being in an exam room where the questions
      keep changing, the clock is speeding up, and everyone
      else has finished. Sometimes the room starts flooding
      or the walls close in.",
    "frequency": "5-6 times per week",
    "duration": "8 months"
  },
  "personality_traits": [
    "worried", "seeking reassurance",
    "catastrophizing", "hyperaware of sensations",
    "needs frequent validation"
  ],
  "resistance_level": "moderate",
  "sample_responses": [
    "What if I try to change the dream and it makes it
     worse? Has that ever happened?",
    "I’m already anxious just thinking about the nightmare.
     I don’t know if I can do this."
  ]
}
```

C. Supplementary Results

This appendix contains figures that support the main text but were deferred to keep the results chapter focused on the primary findings.

C.1. Plan Validity Rate

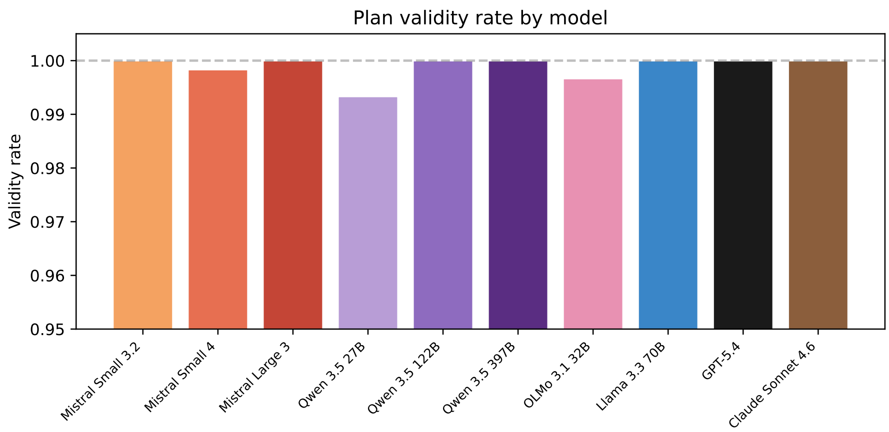

Figure C.1.: Plan validity rate by model, pooled across temperatures and vignettes. All models exceed 99%. The y-axis begins at 0.95 to show the small differences between models.

C.2. BERTScore by Model and Temperature

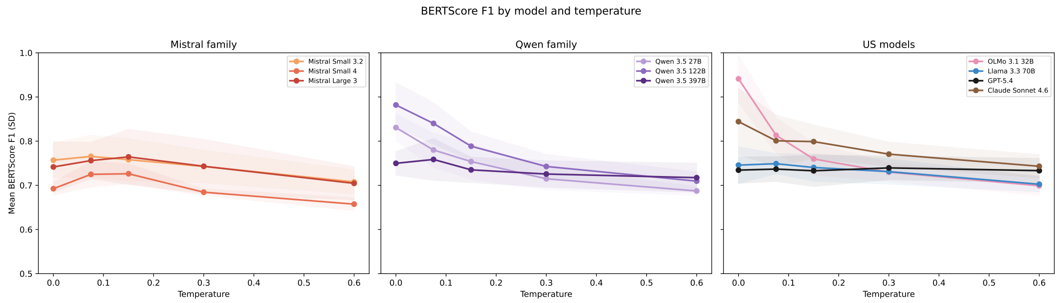

Figure C.2.: Mean BERTScore F1 by model and temperature, grouped by family. The pooled temperature effect is stronger than for Jaccard ($\rho = -0.486$), but BERTScore is insensitive to plan-level differences (section 5.3).

C.3. Vignette Heatmaps

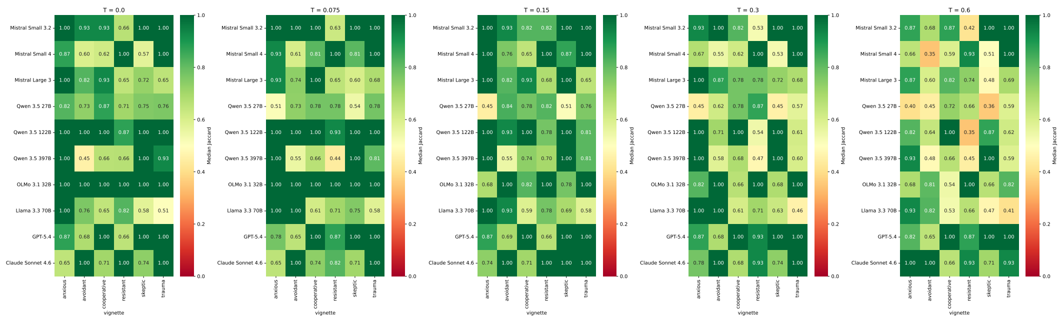

Figure C.3.: Jaccard by model and vignette, one panel per temperature. Some vignettes (e. g. resistant, trauma) show lower consistency across multiple models. The significant vignette effect ($p = 0.007$) is visible as column-wise variation.

C.4. Slice Depth Heatmap

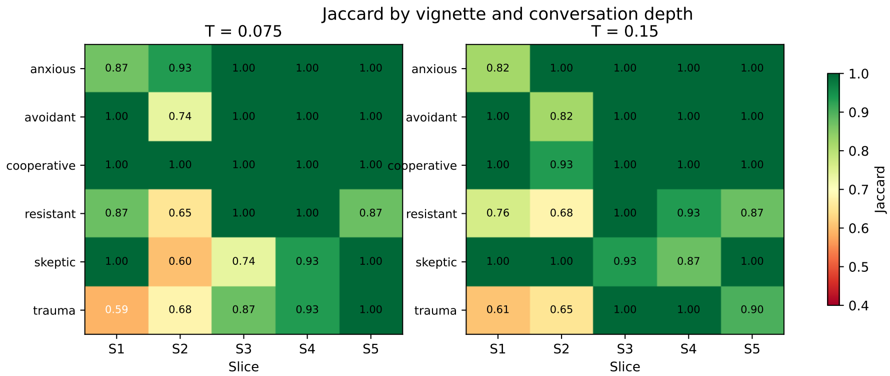

Figure C.4.: Jaccard by vignette and conversation depth for Mistral Large 3, one panel per temperature. Slice 2 shows the lowest consistency across most vignettes, supporting its selection as a conservative evaluation point.